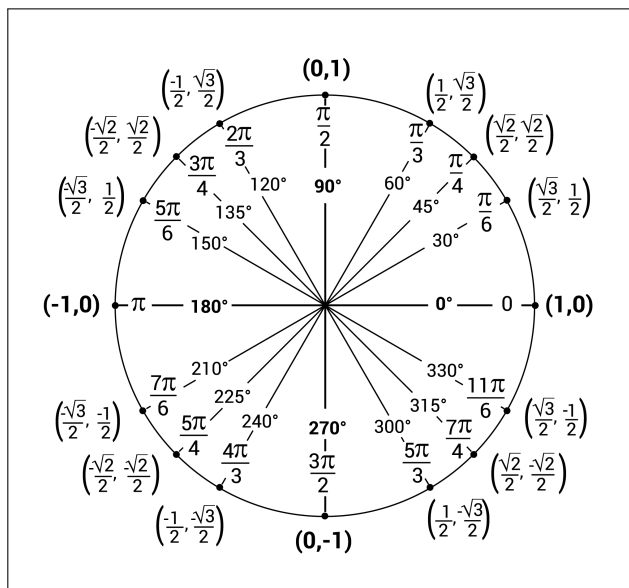


Extra Unit B practice 3

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the final practice worksheets, and the practice problems. The formulas below will be provided on Opportunity B.

Formulas

- $L(x) = f'(a)(x - a) + f(a)$
- MVT: $\frac{f(b) - f(a)}{b - a} = f'(c)$
- Basic derivatives
 - $\frac{d}{dx}(x^n) = nx^{n-1}$
 - $\frac{d}{dx}(e^x) = e^x$
 - $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$
 - $\frac{d}{dx}(\sin x) = \cos x$
 - $\frac{d}{dx}(\cos x) = -\sin x$
 - $\frac{d}{dx}(\tan x) = \sec^2 x$
 - $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$
 - $\frac{d}{dx}(\cot x) = -\csc^2 x$
 - $\frac{d}{dx}(\csc x) = -\csc x \cdot \cot x$
- Derivative rules
 - $\frac{d}{dx}(f(x) \cdot g(x)) = f'(x) \cdot g(x) + g'(x) \cdot f(x)$
 - $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
 - $\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$



B1 - Extrema of functions

Consider the function

$$f(x) = \begin{cases} 2x^3 + 15x^2 & -7 \leq x < -1 \\ 13 & x = -1 \\ (x-7)^{2/3}(x-32) + 145 & -1 < x \leq 15 \end{cases}$$

(a) Use the definition of continuity to show that $f(x)$ is continuous at $x = -1$.

$$\lim_{x \rightarrow -1^-} f(x) = 2(-1)^3 + 15(-1)^2 = 13$$

$$\lim_{x \rightarrow -1^+} f(x) = (-1-7)^{2/3} \cdot (-1-32) + 145 = -4 \cdot -33 + 145 = 13$$

$$f(-1) = 13$$

$$\text{so } \lim_{x \rightarrow -1} f(x) = f(-1) \quad \checkmark$$

(b) Determine all critical points of $f(x)$ (if any).

$$f'(x) = \begin{cases} 6x^2 + 30x & -7 \leq x < -1 \\ \frac{5x-85}{3^3\sqrt{x-7}} & -1 < x < 7 \\ 7 < x \leq 15 \end{cases}$$

$$\frac{\frac{2}{3}(x-7)^{-1/3}(x-32) + (x-7)^{2/3} \cdot 1}{3^3\sqrt{x-7}} = \frac{2(x-32) + 3(x-7)}{3^3\sqrt{x-7}} = \frac{5x-85}{3^3\sqrt{x-7}}$$

$$6x^2 + 30x = 0 \rightarrow 6x(x+5) = 0 \rightarrow x = 0, -5$$

$$\frac{5x-85}{3^3\sqrt{x-7}} = 0 \text{ @ } x=17 \text{ (not in } [-1, 15])$$

$$\text{not in } [-7, -1]$$

$$f(-7) = 2(-7)^3 + 15(-7)^2 = 49$$

$$f(-5) = 2(-5)^3 + 15(-5)^2 = 125$$

$$f(-1) = 13 \leftarrow \text{abs. min}$$

$$f(7) = (7-7)^{2/3}(7-32) + 145 = 145 \leftarrow \text{abs. max}$$

$$f(15) = (15-7)^{2/3}(15-32) + 145 = 77$$

(c) Find the absolute maximum and minimum values of $f(x)$ (if any).

$$\lim_{x \rightarrow -1^-} f'(x) \neq \lim_{x \rightarrow -1^+} f'(x)$$

so $f'(-1)$ DNE

so critical points are:

$$x = -5, -1, 7$$

B2 - The Mean Value Theorem

Consider the function $f(x) = x^3 + e^x - 2\sin(x)$. In each of the following questions, if you rely on any theorems, you must state why those theorems can be used.

- (a) Find an interval on which we know $f(x)$ must have a critical point.

$$f'(x) = 3x^2 + e^x - 2\cos(x)$$

$$f'(0) = 3 \cdot 0^2 + e^0 - 2\cos(0) = -1$$

$$f'(1) = 3 \cdot 1^2 + e^1 - 2\cos(1) = 3 + e - 2\cos(1) > 0$$

So since $f'(x)$ is continuous, by IVT there must be a critical point in $(0,1)$.

- (b) Find another interval, that doesn't overlap with the one from part (a), on which we know $f(x)$ must also have a critical point.

$$f'(-1) = 3 \cdot (-1)^2 + e^{-1} - 2\cos(-1) > 0$$

So again by IVT, there is another critical point in $(-1,0)$

- (c) Show that $f(x)$ has no other critical points than the two in the intervals from parts (a) and (b).

$$f''(x) = 6x + e^x + 2\sin(x)$$

$$f''(x) = 6 + e^x + 2\cos(x) = 0 \text{ has no solutions as } -2 \leq 2\cos(x) \leq 2.$$

f'' is differentiable, so by the corollary to Rolle's Theorem, f'' can have at most 1 root. Then, f' is differentiable, so again by the corollary it can have at most 2 roots (i.e. critical points of $f(x)$).

B3 - Curve sketching

The derivative $f'(x)$ of a function is graphed on the right. The derivative has asymptotes at $x = -1$ and $x = 1$. Suppose we also know the following about $f(x)$:

- $\lim_{x \rightarrow \infty} f(x) = \infty$
- $\lim_{x \rightarrow -\infty} f(x) = \infty$
- $f(0) = -1$
- The x -intercepts of $f(x)$ are 1 and -1 .

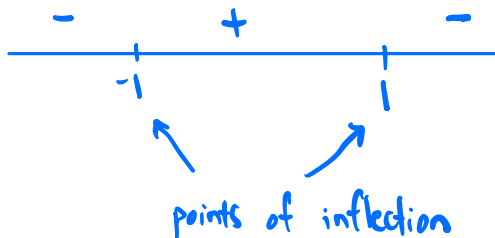
For each of the following questions, you don't need to show any work.

- (a) What are the x -coordinates of all critical points of f . Classify each critical point as a local maximum, local minimum, or neither.

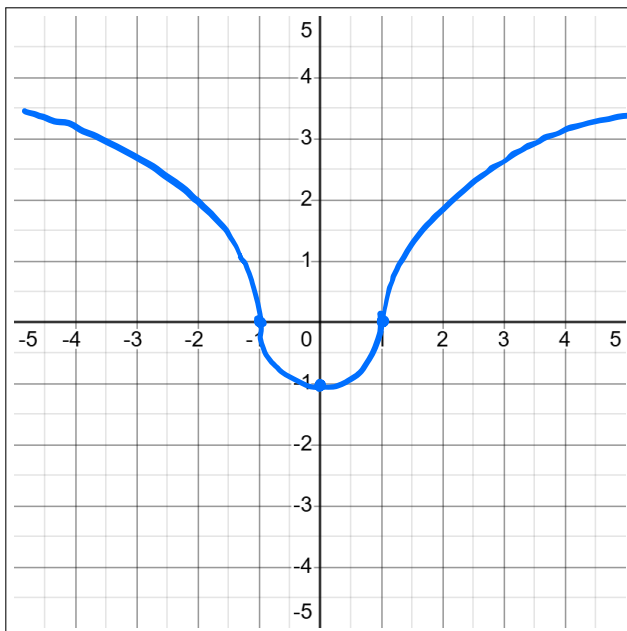
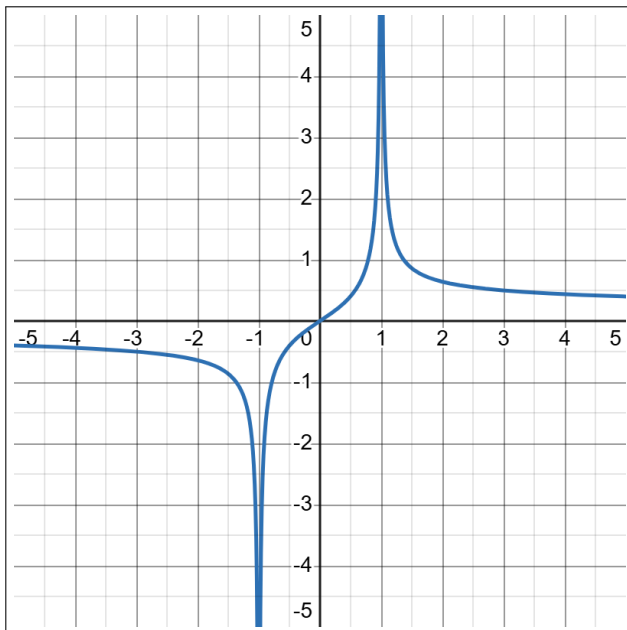
$$x = -1, 0, 1$$



- (b) Construct a second derivative sign chart and identify any points of inflection.

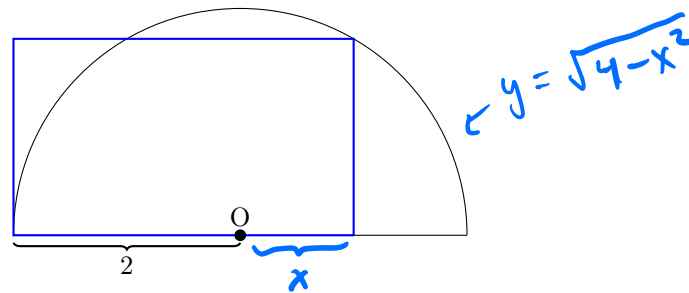


- (c) Use the information given in the problem along with your answers to parts (a) and (b) to sketch an accurate graph of $f(x)$ on the empty pair of axes on the right. Be careful to clearly draw the appropriate intercepts, critical points, concavity, and end behavior.



B4 - Optimization

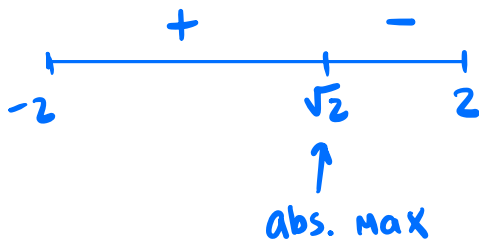
What is the longest perimeter of a rectangle with one side along the base of a semi-circle of radius 2 and with one corner on the arc of the semicircle, as in the figure shown below.



$$P(x) = 2 \cdot (2+x) + 2 \cdot \sqrt{4-x^2} \quad \text{on } [-2, 2]$$

$$P'(x) = 2 + 2 \cdot \frac{1}{2\sqrt{4-x^2}} \cdot (-2x) = 2 - \frac{2x}{\sqrt{4-x^2}} = 0$$

$$\sqrt{4-x^2} = x \rightarrow 4-x^2 = x^2 \rightarrow x = \sqrt{2}$$



$$P(\sqrt{2}) = 2 \cdot (2+\sqrt{2}) + 2 \sqrt{4-(\sqrt{2})^2}$$

$$\boxed{= 4 + 4\sqrt{2}}$$

B5 - Antiderivatives

- (a) Find the most general antiderivative of the following function, on the interval $(0, \infty)$.

$$\begin{aligned} f(x) &= \frac{1 + x^2 + x^5}{\sqrt{x}} \\ &= x^{-\frac{1}{2}} + x^{\frac{3}{2}} + x^{\frac{9}{2}} \\ &= 2x^{\frac{1}{2}} + \frac{2}{5}x^{\frac{5}{2}} + \frac{2}{11}x^{\frac{11}{2}} + C \end{aligned}$$

- (b) Suppose the acceleration of a particle (in m/sec²) at time t (in sec) is given by the function

$$a(t) = 6t - 6$$

for $t \geq 0$.

- (i) Find the velocity $v(t)$ of the particle as a function of time if $v(0) = -9$ m/sec.

$$\begin{aligned} v(t) &= 3t^2 - 6t + C \\ v(0) &= 3(0)^2 - 6(0) + C = -9 \rightarrow C = -9 \end{aligned}$$

$$v(t) = 3t^2 - 6t - 9$$

- (ii) Find the change in position of the particle from $t = 0$ to $t = 4$. Include units in your answer.

$$\begin{aligned} s(t) &= t^3 - 3t^2 - 9t + D \\ s(4) - s(0) &= 4^3 - 3 \cdot 4^2 - 9 \cdot 4 + D - (0^3 - 3 \cdot 0^2 - 9 \cdot 0 + D) \\ &= 64 - 48 - 36 = \boxed{-20 \text{ m}} \end{aligned}$$